



TattvaDrishti

External Project Overview, Technical Architecture and Pilot Proposal

Prepared for: Director, National Cyber Security Research Council (NCSRC)

Prepared by Team ASHTOJ, Keshav Memorial Institute of Technology, Hyderabad

Analyst-Centric Intelligence for AI-Enabled Malign Information Operations

Document purpose

This document presents the current TattvaDrishti prototype, its technical and operational design, responsible-use boundaries, and a proposed pathway toward an independently evaluated agency pilot.

Document status: External project overview

Version 1.0 | July 2026

DOCUMENT GUIDE

Contents

1	Executive Summary
2	National-Security Problem and Operational Need
3	Solution Overview and Design Principles
4	System Architecture and Analytical Workflow
5	Current Prototype Capabilities
6	Batch Processing and Engineering Scale
7	Security, Evidence Governance and Deployment
8	Validation Status, Limitations and Responsible Use
9	Roadmap to an Operational Pilot
10	Proposed Engagement with NCSRC
11	Project Information and Selected References

Executive Summary

TattvaDrishti is an analyst-centric cyber-intelligence and decision-support platform developed by Team ASHTOJ in response to the growing use of generative AI in malign information operations. It is designed for police, cyber-security, intelligence and trust-and-safety teams that must review large volumes of suspicious content while preserving evidence, uncertainty and human accountability.

The platform accepts individual narratives, structured message archives, selected public-channel content and images. It combines multiple independent evidence sources - local semantic-risk reasoning, transformer-based AI-origin estimation, behavioural manipulation indicators and stylometric analysis - and converts them into an explainable analyst case. Each case can retain source identifiers, timestamps, platform and actor metadata, provenance indicators, graph relationships, geographic context, audit history and controlled-sharing rules.

Operational objective Reduce repetitive triage and help analysts reach a defensible assessment faster.	Analytical approach Fuse independent evidence rather than treating one classifier as the final verdict.
Campaign view Connect messages to actors, narratives, regions, URLs and recurring propagation patterns.	Deployment direction Support sovereign/local processing, auditable workflows and standards-aligned sharing.
Core positioning TattvaDrishti is not a fake-news detector or an autonomous enforcement system. It is an evidence-fusing workflow that prioritizes cases, explains signals, reveals relationships and supports controlled human decision-making.	

National-Security Problem and Operational Need

Generative AI lowers the cost of producing fluent, targeted and rapidly adaptable narratives. The operational challenge is not limited to deciding whether a sentence appears machine-generated. Investigators must determine which claims matter, whether manipulation is present, how the content relates to actors or campaigns, what evidence supports the assessment, and what action is proportionate and legally authorized.

Operational gap	Why it matters	TattvaDrishti response
Manual review at scale	Thousands of messages and channel posts must be reviewed under time pressure.	Batch ingestion, prioritization, keyword triage and case filtering.
Single-score detectors	A probability does not explain intent, campaign context, evidence quality or uncertainty.	Multi-signal analysis with visible evidence, disagreement and provenance.
Disconnected cases	Repeated narratives and actor relationships remain hidden across sources.	Graph intelligence across content, actors, narratives, regions and URLs.
Weak evidence handling	Exports can lose source identifiers, timestamps and handling context.	Lineage-preserving schema, fingerprints, audit records and controlled sharing.
Complex specialist tools	Field personnel need a simple result while analysts need detailed drill-down.	Role-specific interfaces with progressive disclosure.

Operational gap	Why it matters	TattvaDrishti response
Human accountability A risk tier is a review priority, not a determination of truth, identity, malicious intent or guilt. High-impact decisions require corroborating evidence, policy authorization and analyst judgement.		

3

Solution Overview and Design Principles

TattvaDrishti converts raw digital content into a structured analyst case. The design connects collection, normalization, multi-signal analysis, campaign correlation, human-readable explanation and controlled sharing in one modular workflow.

Evidence fusion Independent signals remain inspectable and are combined only at the decision-support layer.	Explainability by construction Features and heuristics are recorded during analysis rather than added after a black-box verdict.
Graceful degradation Optional model services can be disabled or temporarily unavailable without collapsing the complete workflow.	Evidence lineage Upstream IDs, timestamps, collections, actors and handling markings remain attached to analytical results.
Sovereign processing Core text analysis can operate locally, supporting sensitive and restricted environments.	Human-in-the-loop operation The interface presents uncertainty, evidence for and against, and a recommended next investigative step.

TattvaDrishti in one operational flow

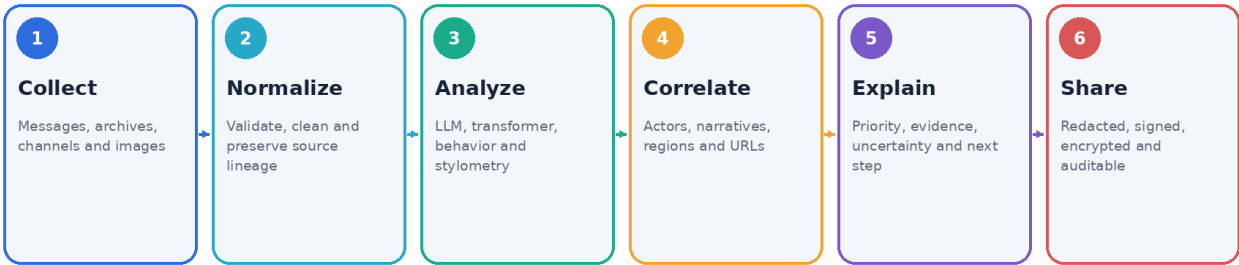


Figure 1. TattvaDrishti operational workflow from collection to controlled sharing.

4

System Architecture and Analytical Workflow

The analytical architecture separates four evidence planes so that no single detector becomes the final verdict. The current prototype uses configurable heuristic weights; the target architecture will learn, calibrate and validate fusion against representative operational data.

Evidence-plane architecture

Independent evidence is preserved and fused; no single detector becomes the final verdict.

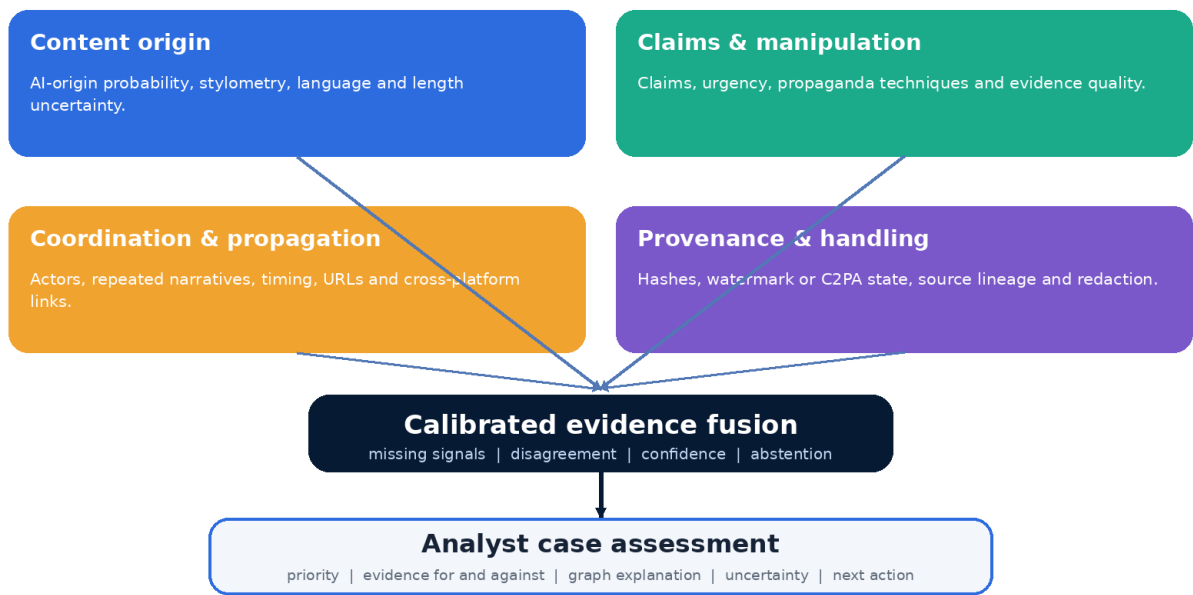


Figure 2. Independent evidence planes feeding a calibrated analyst assessment.

Main request flow

- Intake is validated, normalized and linked to source identifiers, observation time, platform, actor and handling metadata.
- Stylometric and behavioural features are prepared locally; transformer and local-LLM services add AI-origin and semantic-risk evidence when enabled.
- Available signals are fused into a bounded score and operational priority, with controlled reweighting when an optional model is unavailable.
- Provenance markers and fingerprints are checked, and the case is added to an actor-content-narrative-region graph.
- The result, evidence lineage and audit event are persisted; connected dashboards receive a live event.
- The analyst reviews explanations, related cases, keyword filters, graph relationships and controlled-sharing options.

Signal	Prototype weight	Operational role
Local semantic-risk assessment	40%	Contextual danger, manipulation and intent.
Transformer AI-origin estimator	35%	Estimates AI-versus-human origin and supports vectorized inference.
Behavioural analysis	15%	Urgency, calls to action, emotional pressure and contextual cues.
Stylometric analysis	10%	Lexical diversity, entropy, repetition, burstiness and writing patterns.

Current Prototype Capabilities

Capability	Status	What it means today
Manual narrative intake	Implemented	Creates a case from text, language, region, source, platform, actor and tags; persists results and emits a live event.
Batch message archives	Implemented	Accepts versioned JSON plus JSONL/NDJSON uploads with per-record validation, lineage and mixed success/error outcomes.
Public Telegram analysis	Implemented, specialized	Extracts public posts, sanitizes text and returns AI-origin and lexical telemetry.
Reddit/thread normalization	Connector extension	Transforms selected public pages or threads into the common batch schema.
Image analysis	Prototype with external service	Supports selected AI-image, violence, gore, offensive-content, text, QR and image-property checks.
Threat graph	Implemented prototype	Links content, actors, narratives and regions; generates clusters, rankings and propagation indicators.
Geographic risk map	Implemented prototype	Stores normalized case scores as geographic points and visualizes regional distribution.
Threat-intel and SIEM feeds	Implemented prototype	Exposes correlation-oriented payloads for integration with broader security workflows.
Secure sharing	Implemented prototype	Creates redacted, signed and encrypted packages with justification and audit records.
Multilingual validation	Planned	Requires representative Indic and code-mixed datasets, calibration and analyst studies.

Role-specific interfaces

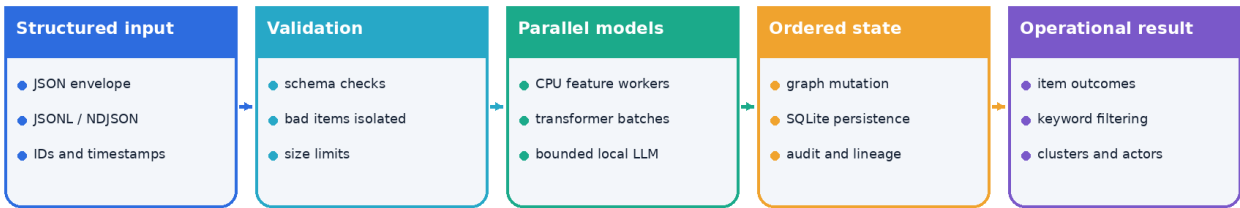
Guided dashboard For police officers, field staff and non-technical personnel. Presents plain-language priority, key findings and the recommended next step.	Analyst dashboard For cyber, intelligence and trust-and-safety analysts. Supports manual and bulk intake, drill-down, graphs, heatmaps, live events and sharing.	Superuser dashboard For administrators and technical supervisors. Supports federation health, validation controls and system-wide monitoring.
--	--	---

Batch Processing and Engineering Scale

The most significant internship extension is the batch-processing path. A complete collection is validated and prepared as one job, model work is vectorized where possible, local-LLM calls are bounded, and graph and storage mutations remain ordered. This converts the prototype from a single-message demonstration into a system that can process real evidence exports while retaining source lineage.

Batch ingestion and parallel analysis

Large evidence exports become validated, searchable and correlated cases while lineage is retained.



~150 messages in approximately 10-15 seconds

Team-reported observation on a specific test setup. It demonstrates batching and parallelization, but is not yet a reproducible benchmark. Throughput varies with hardware, models, message length, warm-up state and local-LLM usage.

Figure 3. Batch ingestion separates parallel model work from ordered graph and persistence operations.

Engineering control	Current default
Default record limit	5,000
Default upload limit	25 MiB
Default transformer batch size	16
Default concurrent local-LLM calls	2

Prototype demonstration

A recent test on the team's setup processed approximately 150 messages in 10-15 seconds after batching and parallelization improvements. This is an engineering observation, not a formal benchmark. A publishable result requires documented hardware, model configuration, message-length distribution, warm-up policy, repeated runs, p50/p95 latency, throughput and failure rate.

7

Security, Evidence Governance and Deployment

TattvaDrishti may process adversarial content, personal data, political speech, sensitive sources and intelligence-derived material. Security, privacy and governance therefore determine whether the analytical result is operationally usable.

Control area	Current or target mechanism	Operational purpose
Data sovereignty	Local model execution, configurable model paths and CPU/GPU operation.	Core text analysis can remain inside agency infrastructure.
Evidence integrity	SHA-256 hashes, fingerprints, signatures, timestamps, source IDs and audit events.	Supports traceability and later verification.
Controlled sharing	Default actor redaction, justification, handling tags, signed packages and encrypted payloads.	Reduces unnecessary disclosure and records accountability.
Graceful degradation	Optional models can be disabled or unavailable without collapsing the complete analysis path.	Supports lightweight nodes and resilient operation.
Interoperability direction	Production schemas should align with STIX 2.1, TAXII 2.1 and TLP 2.0.	Avoids locking agencies into a proprietary case format.

Control area	Current or target mechanism	Operational purpose
Required hardening	OIDC/SAML, MFA, RBAC/ABAC, encrypted storage, file scanning, prompt-injection defenses, retention, backups and change control.	Necessary before an operational pilot.

Evidence sharing model

The current repository includes a tamper-evident federated sharing prototype. For external and operational communication, the recommended production path is standard encryption in transit and at rest, signed packages, interoperable cyber-threat-intelligence objects, policy-based access, acknowledgement, correction and revocation records, and append-only audit logs. Blockchain terminology should not obscure the simpler security and governance requirements of inter-agency exchange.

Provenance is not truth

A valid signature, watermark or C2PA record can establish aspects of an asset's history and integrity. It does not prove that the content is factually correct. The absence of provenance also does not, by itself, prove manipulation.

8

Validation Status, Limitations and Responsible Use

Limitation	Current implication	Required response
Model validity	Current detectors and fixed fusion can fail under paraphrase, translation, short text, unseen generators and domain shift.	Benchmark multiple methods, calibrate by subgroup and support abstention.
Performance evidence	The 150-message test is not a reproducible load benchmark.	Publish configuration, repeated runs, latency percentiles, throughput and failure analysis.
Database and jobs	SQLite and synchronous responses are suitable for a prototype, not sustained operations.	Move to PostgreSQL, a durable asynchronous queue and object storage.
Connector coverage	Public-source connectors depend on platform availability, terms and rate limits.	Use curated feeds, formal agreements and connector-health monitoring.
Provenance	Current watermark and fingerprint checks are prototype indicators.	Add a C2PA parser, trust policy, original-byte preservation and transformation testing.
Multilinguality	English-oriented thresholds do not transfer directly to Indic and code-mixed content.	Build gold sets and calibrate by language, script, platform and length.
Graph scale and causality	Correlation does not establish intent; NetworkX is not yet benchmarked for large temporal graphs.	Evaluate unseen campaigns, include organic controls and migrate to scalable graph analytics.
User validation	Informal feedback identified navigation and value-proposition challenges.	Conduct structured studies with police, intelligence and SOC personnel.

Responsible-use boundary

The platform should support lawful, authorized and proportionate investigations. Monitoring, retention, identity resolution and inter-agency sharing require explicit legal authority, policy controls, oversight and documented analyst accountability.

Roadmap to an Operational Pilot

Phase	Priority work	Outcome
Phase 1 - Reliability and security	Authentication and roles, input/file hardening, structured logging, secrets management, encrypted storage, dependency and model versioning.	Pilot-ready engineering baseline.
Phase 2 - Data and scale	PostgreSQL, durable job queue, object storage, connector registry, deduplication, observability and formal load tests.	Repeatable high-volume ingestion.
Phase 3 - Evidence quality	Modern origin baselines, retrieval-grounded claim verification, propaganda-span detection, calibrated fusion, multilingual gold sets and red-team attacks.	Scientifically defensible assessments.
Phase 4 - Campaign intelligence	Temporal and multiplex graph models, cross-platform entity resolution, narrative migration, C2PA and media provenance.	Campaign-level investigation.
Phase 5 - Operational integration	Analyst studies, STIX/TAXII/TLP exchange, SIEM/SOC playbooks, legal-policy controls, correction/revocation and independent evaluation.	Agency deployment and governance.

Pilot success criterion

A successful pilot should demonstrate that trained users can triage representative cases faster, with equal or better evidence quality, fewer duplicate alerts and appropriate rejection of incorrect model recommendations.

Proposed Engagement with NCSRC

TattvaDrishti would benefit from independent cyber-security, operational and governance review before any agency pilot. Collaboration with NCSRC could help convert the current prototype into an evaluated, responsible and interoperable system.

Technical architecture review Assess local deployment, ingestion, model isolation, evidence handling, attack surface and integration boundaries.	Evaluation design Define representative datasets, adversarial tests, multilingual calibration, load-testing methodology and analyst-study protocols.
Operational workflow review Validate how police and intelligence users receive priorities, evidence, uncertainty, graph explanations and recommended next steps.	Governance and standards alignment Review access control, privacy, retention, audit, sharing and alignment with CTI and provenance standards.
Pilot pathway Identify a bounded, lawful and non-production evaluation environment with clear success criteria and oversight.	

Requested outcome

An expert-reviewed pilot plan covering technical hardening, scientific validation, operational usability, security governance and measurable deployment criteria.

Project Information and Selected References

Field	Value
Project	TattvaDrishti - Analyst-Centric Intelligence for AI-Enabled Malign Information Operations
Team	Team ASHTOJ
Institute	Keshav Memorial Institute of Technology, Hyderabad
Origin	Smart India Hackathon 2025 project; extended through the AICTE Cyber Security Internship 2025-2026
Repository	https://github.com/shoaib2000857/TattvaDrishti-AICTE-Internship
Implementation baseline	Repository and project documentation reviewed through July 2026
Document status	External project overview and pilot proposal

Selected standards and technical foundations

- OASIS STIX 2.1 and TAXII 2.1 for structured cyber-threat-intelligence exchange.
- FIRST Traffic Light Protocol 2.0 for information-sharing handling and dissemination.
- NIST AI Risk Management Framework and Generative AI Profile for risk governance.
- OWASP guidance for securing applications that use large language models.
- Coalition for Content Provenance and Authenticity (C2PA) technical specification.
- Evidence-centered claim-verification and misinformation-diffusion research used to shape the validation roadmap.

Closing statement

TattvaDrishti does not replace the analyst. It gives the analyst a faster, connected and more defensible view of the evidence.